

Title:**Sentiment Insights – A Social Media Platform with Real-Time Sentiment Analysis****Introduction:**

In today's digital world, communication through social media platforms has become very common. However, understanding the emotional tone behind user-generated content is often difficult. This project, *Sentiment Insights*, aims to bridge this gap by integrating sentiment analysis into a content-sharing platform, enabling users to understand the emotions behind text and images in real time.

Objective:

The main objective of this project is to develop a web-based application that allows users to share text and images while automatically analyzing and displaying their sentiment. The system helps users gain insights into the emotional tone of posts and comments, thereby improving communication and interaction.

Problem Statement:

Existing social media platforms allow users to share content but do not provide tools to analyze the emotional tone of messages. This lack of sentiment understanding can lead to miscommunication and ineffective interactions. Therefore, there is a need for a system that combines content sharing with real-time sentiment analysis.

Proposed System:

The proposed system is a web application built using Flask that enables users to:

- Register and log in securely
- Post text and image content
- View posts from other users
- Comment on posts

The system integrates:

- **Text Sentiment Analysis** using NLTK's VADER to classify content as positive, negative, or neutral
- **Image Emotion Detection** using a TensorFlow model to identify emotions such as happy, sad, angry, etc.

All posts and comments are analyzed in real time, providing instant feedback to users.

Methodology:

- Data is collected from user inputs (text and images)
- Text is processed using NLP techniques and analyzed using VADER
- Images are processed using a CNN-based deep learning model for emotion recognition
- Results are displayed on the user interface along with the content

Technologies Used:

- **Frontend:** HTML, CSS, Bootstrap
- **Backend:** Python (Flask)
- **Database:** SQLite
- **Libraries/Tools:** NLTK, TensorFlow, SQLAlchemy

Modules:

1. User Management (Login/Register)
2. Content Management (Post & View Content)
3. Sentiment Analysis (Text & Image)
4. Comment System
5. Database Management

Advantages:

- Provides real-time sentiment feedback
- Improves communication clarity
- Helps in content moderation
- Enhances user engagement

Applications:

- Social media platforms
- Customer feedback analysis
- Online review systems
- Mental health and emotional monitoring

Conclusion:

The *Sentiment Insights* project successfully integrates sentiment analysis with a social media platform. It helps users better understand the emotional tone of interactions, leading to more meaningful and positive communication.

Future Enhancements:

- Multi-language sentiment analysis
- Advanced deep learning models for better accuracy
- Real-time chat sentiment tracking
- Deployment on cloud platforms